

Cosmetic Product Safety Assessment of
Capsule Blood

This safety assessment relates to the formulation described below. If the information below is incorrect, please amend and resubmit for reassessment.

KIMYON INDUSTRIAL (HK) COMPANY LIMITED
UNIT B, 20/FL., FORD GLORY PLAZA,
37 WING HONG STREET , CHEUNG SHA WAN,
KOWLOON, HK, P.R. CHINA

PRODUCT FORMULATION

The chemical names shown below refer to the raw materials used to formulate this product. The identity of the raw materials is not necessarily based on the International Nomenclature of Cosmetic Ingredients (INCI) and does not represent the INCI listing that must be shown on the product label and is for assessment purposes only.

Chemical Name	Conc	% Active	Product	CAS No	Einecs No
PEG-400	94.2995	100	94.2995	25322-68-3	203-473-3
AQUA (WATER)	3.5	100	3.5	7732-18-5	231-791-2
SORBITOL	1.5	100	1.5	50-70-4	200-061-5
SUCRALOSE	0.1	100	.1	56038-13-2	259-952-2
POTASSIUM SORBATE	0.1	100	.1	24634-61-5 / 590-00-1	246-376-1
CI 16035 (FD&C RED NO. 40)	0.5	100	.5	25956-17-6	247-368-0
CI 42090 (FD&C BLUE NO. 1)	0.0005	100	.0005	3844-45-9	223-339-8

LABELLED WARNINGS & INSTRUCTIONS OF USE

CONSUMER EXPOSURE

Product Class: Halloween Blood Capsules
IFRA Product type: Mouthwash, including Breath Sprays
IFRA Category: Category 6
Targeted Population: Children 14 years of age 50.4kg (Mean)
Amount per application/g: 0.740
Skin Surface Area of Application/cm²: 216.800
Total Amount applied per day/g: 0.74
Estimated Daily Exposure mg/kg/day: 12.3
Amount Per Unit Area of Skin per day mg/cm²/day: 3.410
Retention factor: 0.50
Exposure Time Neat: 30 seconds (1 g ingested-COLIPA, 1993 93/067)
Exposure Time Dilute: Not applicable
Exposure time Solvent Inhalation: Not Applicable
Exposure time Aerosol Inhalation: Not Applicable
Number of applications per day: Once a day
Physical form: Liquid
Part of body exposed to undiluted: Mouth

This product has been assessed taking into account that it will be used by teenagers aged 14 years and older.

STABILITY OF COSMETIC PRODUCT

It is assumed that the responsible person has selected all pertinent criteria required to evaluate the stability of this cosmetic product during reasonable foreseeable conditions of storage. The stability report provided by the supplier and based upon the conclusions made therein, this cosmetic product appears to be stable under reasonably foreseeable storage conditions.
53-week stability test data at -18°C, 5°C and 40 °C as well as cycle test (2023.01.11) has been provided by applicant. Based on the physiochemical data as well as microbiological data stated, the product is stable at least over a period of 2 years.

MICROBIOLOGICAL QUALITY

To comply with the Guidelines on the Microbiological Quality (SCCS/1564/15), the following maximum limits apply:

Category 1: Products specifically intended for children under 3 years, eye area and mucous membranes.

TVC:- 100 cfu/g or ml in 0.5 g or ml of the product.

Pseudomonas aeruginosa, Staphylococcus aureus and Candida albicans must not be detectable in 0.5 g or ml of the cosmetic product

Category 2: Other cosmetic products.

TVC:- 1000 cfu/g or ml in 0.1 g or ml of the product

Pseudomonas aeruginosa, Staphylococcus aureus and Candida albicans must not be detectable in 0.1 g or ml of the cosmetic product

The microbiological specifications for the product have been supplied and based upon the conclusions therein, meet the Industry requirements specified in the SCCS's Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 8th Revision, Section 4-4.2

The preservative challenge test results for this product have been supplied and based upon the conclusions made therein appear to meet the industry requirements specified in the SCCS's Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 8th Revision, Section 4-4.3

Microbiological contamination test report (2024.02.09) has been provided by applicant, demonstrating hygiene condition of product complying to limits under Cat. 2 stated above.

Preservative challenge test report (2022.08.04) has been provided by applicant, demonstrating performance of product formulation against contamination from index micro-organisms.

PACKAGING INFORMATION/COMPATIBILITY

It is assumed that the responsible person has identified the most applicable testing required to determine the packaging stability and its interaction with the cosmetic product contained within it. Taking into consideration the information supplied to the assessor, there appears to be no immediate health concern due to the characteristics of packaging materials in direct contact with the final product.

Testing data on capsule material has been provided (2024.02.22 & 2023.12.06), indicating the components not containing heavy metals at levels of concern, with lead, cadmium, mercury and chromium (VI) below 100 ppm requirement as laid down under Directive 94/62/EC for packaging and packaging waste and meeting phthalates content as per Annex XVII of the REACH Regulation (EC) no. 1907/2006.

SERIOUS / UNDESIRABLE EFFECTS

The supplier has confirmed that no undesirable effects or serious undesirable effects of this cosmetic product have been reported or, where relevant, cosmetic products with a similar formulation.

HUMAN STUDIES

No existing studies from human volunteers were provided at the time of assessment.

FRAGRANCE COMPOSITIONS

This formulation does not contain a synthetic fragrance and therefore a fragrance safety evaluation as per IFRA code of practice is not applicable to this product.

PRESENCE OF NANOMATERIALS

The supplier has confirmed that this cosmetic product does not contain any nanomaterials that are known to them within the meaning of the definition as stated in Cosmetic Product Regulation (EC) No 1223/2009.

IMPURITIES /TRACES/ PROHIBITED SUBSTANCES

Where the specification is provided it is noted that this product does not contain any impurities at levels likely to cause harm to the user

TOXICOLOGICAL & REGULATORY REVIEW

A mixture of binder, solvent, humectant, sweetener, preservative and colorant to give fake blood, encapsulated in a capsule for disguise purpose by teenager 14 years of age or above. Repeated exposure on the face within the proximity of the eyes and body parts as well as short term exposure in oral cavity are expected. Accidental exposure to eye and ingestion during normal use is anticipated but not expected to be a systemic one. The preservative and colorants are approved for use in cosmetics under Cosmetic Regulation (EC) No. 1223/2009 and within the restrictions of use stated if any, while none of the ingredients is on the prohibition list of the regulation. Cross reactions among the ingredients are unlikely to happen based on a review of the chemical compositions of the ingredients. Most of the ingredients being used are not dermally irritating or sensitizing nor acute oral or dermally toxic. Evaluation of the safety of specific concerned ingredients is stated as follows:

PEG-400 has been used as binder. With a NOAEL of 1100 mg/kg bw/day from repeated dose toxicity study in rats, an MoS of over 100 has been yielded, being the lowest determined from this product formulation. Teenagers with mean body weight of 50.4kg would ingest 6.9 mg/kg body weight/day of PEG-400, assuming that half of one capsule of 0.74g was ingested. The Acceptable Daily Intake (ADI) of polyethylene glycol is up to 10 mg/kg body weight/day (Joint FAO/WHO Expert Committee on Food Additives (JECFA) 1979). In considering the occasional use of this Halloween product, this intake is not therefore considered to present risk to health under normal conditions of use of the product.

Where a NOAEL is available for a chemical ingredient that is considered as a toxicological concern, the Margin of Safety (MoS) has been calculated as greater than 100 taking into consideration any known data on dermal absorption and bioavailability. It is generally accepted that the MoS should be a least 100 to declare a substance safe for use in a finished product and the safety of this formulation is further supported by this uncertainty factor.

If used as directed, use of this product should be uneventful.

Health effects of the product as supplied on the skin

The formulation as supplied may cause slight skin irritation especially if exposure is prolonged and/or repeated.

Repeated exposure to the formulation as supplied is unlikely to produce allergy by skin contact.

Exposure to this product is unlikely to result in phototoxic effects.

Unlikely to cause damage to internal organs following absorption through the skin.

Health effects of the product as supplied on the eye

Accidental exposure of the eye to the formulation as supplied may result in slight eye irritation.

Health effects following ingestion of the product as supplied

The formulation as supplied if swallowed may cause slight, transient irritation to the mouth and upper digestive tract.

Health effects of inhaling the product

Inhalation is an unlikely route of exposure

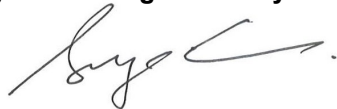
Overall Assessment Conclusion

The ingredients are legally permitted as per Cosmetic Regulation (EC) No. 1223/2009 and its amendments and the safety assessment has been carried out in accordance to this regulation. They must comply with the relevant purity standards for cosmetic ingredients. It is assumed that these ingredients do not contain any undisclosed impurities/contaminants that would affect the conclusions reached. The product must be manufactured in accordance with EU Guidance on Good Manufacturing Practice.

Under normal or reasonably foreseeable conditions of use, a product made to this formulation is unlikely to produce an abnormally high number of adverse reactions. The product will give users the level of safety they can reasonably expect when used as directed.



Toys Toxicological Safety Assessor



Suzanna Chan BSc, MSc Appl Tox, PhD, CChem, FRSC, UKRT, ERT

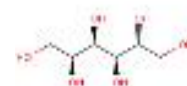
2/F Garment Centre, 576 Castle Peak Road, Kowloon, Hong Kong SAR, China **Date:** 19 Apr 2024

Capsule Blood

TOXICOLOGICAL PROFILE OF SUBSTANCES

Chemical Substance: PEG-400			
EU INCI NAME:PEG-400			
CAS: 25322-68-3		Function: Humectant/Solvent/Gel former	
EINECS 203-473-3			
Appearance: Liquid		Boiling Point: >250°C	
Water Solubility: Miscible			
Cosmetic Regulatory Summary:			
EU Cosmetics Status:		Not controlled	
Saudi Cosmetics Status:		Not controlled	
US Cosmetics Status:		Not controlled	
Regulatory Summary:			
EU Classification:		Not classified	
GHS Classification:		Not classified	
REACH Annex XVII controlled:		Not Controlled	
REACH SVHC Candidate List:		Not Controlled	
Systemic Exposure Dosage / Margin of Safety:			
SED Adult mg/kg bw/day: 0.489	MoS - Adult 60kg: 2246.5	NO(A)EL mg/kg bw day: 1100	
SED Child mg/kg bw/day: 1.759	MoS - Child 16.7kg: 625.2	NOAEL test method:	Subchronic oral study in rat
SED Baby mg/kg bw/day: 4.979	MoS - Baby 5.9kg: 220.9		
Toxicological Summary:			
Cosmetic Functions: Emulsifying / Surfactant /Solubilizing Agent. PEG 400 (trade name for a Polyethylene glycol (PEG) mixture with about 90 units in the chains). Low potential to cause skin or eye irritancy and allergenic reaction. Not acutely toxic. Very little toxicity data are available for the PEGs (CIR Compendium 2009). In general, PEGs have a low level toxicity whether the exposure is oral or dermal. No evidence of mutagenicity, carcinogenicity, or reproductive and developmental toxicity. In burns patients, PEGs were mild irritant/ sensitisers and there was evidence of nephrotoxicity. PEGs are capable of penetrating injured skin and mucosa with compromised barrier function. Repeat dose toxicity of PEGs and their derivatives is very low, with kidney and liver being identified as target organs in animal studies at extremely high exposure levels, which exceeds the recommended levels in testing guidelines. There are few reported cases of sensitisation with PEGs but clinical studies have not shown any significant effects. The overall weight of evidence suggests that this PEG compound is not genotoxic and there is no indication of carcinogenicity potential. There was no evidence for adverse effect on reproduction and development reported with studies on PEGs and their derivatives. Some monoalkyl ether glycols are known reproductive toxicants; however, the available information on the manufacturing process of PEG and the deruvatives suggest that these compounds would not likely be present as impurities. The lowest NOAEL reported in oral repeat dose studies with PEGs was 1100 mg/kg bw/d (Fruijtier-Polloth 2005; Toxicology 214: 1-38). The dermal penetration and absorption of PEG-4 has been investigated. The average total absorption of PEG-4 from a leave-on formulation in intact and damaged skin are 8.42% and 33.72% respectively, while the average total absorption of PEG-4 from a rinse-off formulation in intact and damaged skin are 0.37% and 2.30% respectively.			
Reference: Amended Safety Assessment of Triethylene Glycol and Polyethylene Glycols, CIR, 2010.			

Chemical Substance: AQUA (WATER)			
EU INCI NAME:AQUA			
CAS: 7732-18-5		Function: Solvent	
EINECS 231-791-2			
Appearance: Liquid		Melting Point: 0	
Water Solubility: highly soluble		Boiling Point: 100	
Cosmetic Regulatory Summary:			
EU Cosmetics Status:		Not controlled	
ASEAN Cosmetics Status:		Not controlled	
Saudi Cosmetics Status:		Not controlled by Saudi legislation	
US Cosmetics Status:		Not controlled	
Canadian Cosmetics Status:		Not controlled	
Regulatory Summary:			
EU Classification:		Unclassified	
GHS Classification:		Unclassified	
REACH Annex XVII controlled:		Not Controlled	
REACH SVHC Candidate List:		Not Controlled	
Systemic Exposure Dosage / Margin of Safety:			
SED Adult mg/kg bw/day: 0.215	No NOAEL Available	NO(A)EL mg/kg bw day: -	
SED Child mg/kg bw/day: 0.775	No NOAEL Available	NOAEL test method:	-
SED Baby mg/kg bw/day: 2.194	No NOAEL Available		
Toxicological Summary:			
Function : Solvent. Simply water unlikely to cause irritation, allergy or harm. Used in many cosmetic products as a solvent and necessary to sustain biological life. The source of water should be known, monitored to GMP and either a deionised or high purity grade free from toxins, pollutants and bacteriological contamination should be used in cosmetic products.			



Chemical Structure:

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled
 ASEAN Cosmetics Status: Not controlled
 Saudi Cosmetics Status: Not controlled by Saudi legislation
 US Cosmetics Status: Not controlled
 Canadian Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Not classified
 GHS Classification: Not classified
 REACh Annex XVII controlled: Not Controlled
 REACh SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.074	MoS - Adult 60kg: 101351.3	NO(A)EL mg/kg bw day: 7500
SED Child mg/kg bw/day: 0.265	MoS - Child 16.7kg: 28209.4	NOAEL test method: 2-year dietary study; rat
SED Baby mg/kg bw/day: 0.752	MoS - Baby 5.9kg: 9966.2	

Toxicological Summary:

Cosmetic Functions : Humectant / Plasticiser /Skin Conditioning / Flavoring Agent / Fragrance Ingredient. Sorbitol is a hexahydric sugar alcohol that occurs naturally in plants and animals. It is widely used in the food industry as an ingredient in many food products and medically, sorbitol is used in parenteral nutrition as a source of carbohydrate. It has long history of use as food ingredient/additive and there have been no indications of significant harmful effects reported. Sorbitol is hygroscopic. About 80% of an orally administered dose is absorbed. Once absorbed, it is rapidly metabolised by the liver to several intermediates, but ultimately to glucose. Sorbitol is well utilised by the body as an energy source, and consequently, very little of an oral dose is eliminated in the urine and faeces. The only consistent adverse effect found in animals or humans is mild gastrointestinal effects (GI) such as gas, bloating, abdominal discomfort, and diarrhoea.

Acute toxicity: Sorbitol was practically non-toxic in acute oral studies in laboratory animals (LD50: 7.1 to 25 g/kg bw) and short-term toxicity studies in humans demonstrates that it does not cause any adverse effect other than mild gastrointestinal effects even at high dose levels of between 10-50 g.

Irritation: Non-irritating. There is no data available indicating that sorbitol has skin, eye (mucous membrane) or respiratory tract irritation potential.

Sensitisation: Non-sensitising. No data available in animals and sorbitol is rarely associated with elicitation of allergic dermatitis in humans including occupational exposure.

Mutagenicity/Genotoxicity: Sorbitol is not genotoxic in a wide range of in vitro and in vivo mutagenicity tests.

Repeat dose toxicity: No toxic effects were noted in rats when 5% sorbitol was given in the diet for 3 months corresponding to 5 g/kg bw/day. Administered intravenously to rabbits, no effect on blood chemistry, body weight, survival, or organs at doses of 2.5, 5.0, or 10 mL/kg of a 20% sorbitol solution (500 mg/kg bw/d, 1000 mg/kg bw/d, and 2000 mg/kg bw/d) given daily for 3 months. Sorbitol was fed to weanling rats for two years at dietary levels of 0, 1, 5, or 10%. The two highest dose levels were supplemented with casein to maintain a 25% dietary protein level. A few of the treated animals had diarrhoea during the first 2 months at the 10% feeding level (12 g/kg bw/day). No toxicological effects were reported; the NOAEL was > 12000 mg/kg bw/d. In another 2-year chronic toxicity study, rat fed 20% sorbitol in the diet (~7.5 g/kg bw/d) had enlarged cecum, lower body weight gain, absolute thyroid weight and increased incidence of adrenal medullary hyperplasia. There is no evidence for reproduction, teratogenicity or developmental toxicity potential shown with sorbitol in numerous studies conducted in laboratory animals even at high doses of 1600 – 7800 mg/kg bw/d. Treatment-related tumours were not reported in the chronic toxicity studies which indicates that sorbitol lacks carcinogenicity activity.

Photo-induced toxicity: Non-phototoxic.

Other data (human): Sorbitol causes laxation because of its osmotic effects and poor absorption by the small intestine which results in increased transit time through the colon. In humans, the laxative threshold is 10-20 g of sorbitol liquid and 50 g of crystalline sorbitol; a study in 86 volunteers at a dose level of 25 g/day in two doses did not cause laxation. According to the results of dietary survey conducted by the US National Research Council (NRC), the average daily dietary intake of sorbitol ranges from 770 mg/day in babies to ~31g/day in adults or a maximum of 55.4 g/day.

Dermal / percutaneous absorption: No data available. However, it is assumed that the absorption through the skin would be lower than that of the oral routes; thus, the oral absorption value of 80% is used as conservative estimate.

Chemical Substance: SUCRALOSE

EU INCI NAME: SUCRALOSE

CAS: 56038-13-2

EINECS 259-952-2

Function: Sweetener

Log Kow: -0.51

Cosmetic Regulatory Summary:

EU Cosmetics Status: Not controlled

Regulatory Summary:

EU Classification: Unclassified
 GHS Classification: Not classified
 REACh Annex XVII controlled: Not Controlled
 REACh SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.006	MoS - Adult 60kg: 243243.2	NO(A)EL mg/kg bw day: 1500
SED Child mg/kg bw/day: 0.022	MoS - Child 16.7kg: 67702.7	NOAEL test method: 26-week oral feeding study
SED Baby mg/kg bw/day: 0.062	MoS - Baby 5.9kg: 23918.9	

Toxicological Summary:

Cosmetic function: Skin conditioning. Sucralose is a non-nutritive chlorinated derivative of sucrose that is approved as an artificial sweetener in many countries including the USA and UK. The chlorination gives stability to the molecule which is not metabolised in the body and hence has no calorific value when eaten.

Sucralose is listed in the Foods Chemicals Codex as a flavor enhancer, listed in the USP/NF, an inactive ingredient for approved drugs, used in oral, sublingual, and buccal drugs and can function as a sweetening agent in pharmaceuticals.

Acute toxicity: Oral LD50>16 g/kg in mice and rats

Sensitization: Not an irritant or sensitizer (HRIPT with 50 subjects / lip balm containing 0.6% sucralose)

Repeated toxicity: Sucralose related bodyweight gain effects were investigated in groups of Sprague-Dawley rats (20/sex/group) dosed by gavage with 0-3000 mg/kg bw/day for 26 weeks. Food intake for males dosed with 3000 mg/kg bw/day was 3.9% greater than controls. Interestingly, the adjusted mean body weight gain of males dosed with 3000 mg/kg bw/day was significantly decreased (4.6%; p = 0.035) compared to controls. The NOEL for the bodyweight gain effect observed in this study was determined to be 1500 mg/kg bw/day

Reprotoxicity: not teratogenic or neurotoxic and no effect on reproduction, insulin secretion and carbohydrate metabolism. Sucralose was found to induce a reduction in thymus weight in rats dosed orally with 3000 mg/kg bw/day. The NOEL for immunological endpoints was 750 mg/kg bw/day.

Carcinogenicity: The carcinogenicity of sucralose has been tested in a study where groups of Charles River CD-1 mice (52/sex/group) were fed 0, 0.3, 1 and 3% sucralose in the diet for 104 weeks. During the treatment period the mean body weight gain of mice dosed with 3% sucralose was significantly reduced compared to controls, even though food consumption was normal. A significant decrease in erythrocyte count was also detected in females dosed with 3% sucralose. No evidence of treatment related neoplasia was detected. The dietary NOEL was determined to be 1% sucralose (equivalent to 1500 mg/kg bw/day)

Mutagenicity: negative in vitro (Ames test on S. typhimurium strains TA1535, TA1537, TA1538, TA98, TA100, with and without metabolic activation, 0.16-10 mg/plate and human peripheral lymphocyte assay on human lymphocytes at 8, 40, and 200 µg/ml). Not mutagenic in vivo (micronucleus test; mice (5/sex/group) were given a single dose (2g or 10g/kg bw) by gavage, and micronuclei were evaluated 24, 48, or 72 h after dosing).

CIR: safe as used (up to 0.6% in leave-on products and up to 1.2% in rinse-off products)

References:

CIR. Safety Assessment of Monosaccharides, Disaccharides, and Related Ingredients as Used in cosmetics. 2014
 Safety Assessment of Monosaccharides, Disaccharides, and Related Ingredients as Used in Cosmetics, CIR, 2019.
 NICNAS. Full Public Report SPLENDA/SUCRALOSE. File No: EX/43 (NA/944). 2003

Chemical Substance: POTASSIUM SORBATE

EU INCI NAME:POTASSIUM SORBATE

CAS: 24634-61-5 / 590-00-1

EINECS 246-376-1

Function: Preservative

Appearance: White Granular

Melting Point: 210 °C (Decomposes on heating.)

Log Kow: Data not available.

Boiling Point: Data not available.

Water Solubility: 1,400 g/l at 20 °C (Slightly soluble)

Vapour Pressure: Data not available.

Cosmetic Regulatory Summary:

- EU Cosmetics Status: Approved preservative
- Saudi Cosmetics Status: Permitted preservative - all products.
Max conc 0.6% as a preservative.
- US Cosmetics Status: CIR-Safe up to 7%
- Canadian Cosmetics Status: Controlled

Regulatory Summary:

- EU Classification: R36
- GHS Classification: Eye irrit. 2
- REACH Annex XVII controlled: Not Controlled
- REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.006	MoS - Adult 60kg: 405405.4	NO(A)EL mg/kg bw day: 2500	
SED Child mg/kg bw/day: 0.022	MoS - Child 16.7kg: 112837.8	NOAEL test method:	JEFCA identified NOAEL.
SED Baby mg/kg bw/day: 0.062	MoS - Baby 5.9kg: 39864.8		

Toxicological Summary:

Function: Preservative. As supplied, it is classified as irritating to eyes(R36).
H335 May cause respiratory irritation.A preservative with a good history of safe use. Widely used to preserve plant extracts and also food E202. Permitted at up to 0.6% as acid in the EU. Has optimal function at low range of pH 4 - 6. FEMA/GRAS Approved, FEMA Number 2921.
CIR expert panel concludes this is safe at the present uses and concentrations in a cosmetic product (up to 7%). ADI 25 mg/kg. Subchronic Toxicity Data:
Oral Study (3months, rat): LOEL = 5 % in diet (target organ effects: kidney), NOEL = 2% in diet ; Oral Study (3 months, dog): NOEL = 2 % in diet (highest dose tested)
Reproductive Toxicity Data: Oral Study (rat): NOEL for teratogenicity = 340 mg/kg/day (highest dose tested) Oral Study (mouse): NOEL for teratogenicity = 460 mg/kg/day (highest dose tested);Mutagenicity/Genotoxicity Data: Salmonella typhimurium assay (Ames test): Negative (+/- activation); Mitotic recombination (Saccharomyces cervisiae) assay: Negative (+/- activation. JEFCA identified a NOAEL of 2500mg/kg bw/day.

Chemical Substance: CI 16035 (FD&C RED NO. 40)

EU INCI NAME:CI 16035

CAS: 25956-17-6

EINECS 247-368-0

Function: Colour

Appearance: powder

Melting Point: 349.8

Log Kow: -0.55

Water Solubility: 4.2

Cosmetic Regulatory Summary:

- EU Cosmetics Status: Approved colour all products
- Saudi Cosmetics Status: Permitted colour field 1 All products
- US Cosmetics Status: Generally, including cosmetics 74.2340 including the area of the eye

Regulatory Summary:

- EU Classification: Unclassified
- GHS Classification: Unclassified
- REACH Annex XVII controlled: Not Controlled
- REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

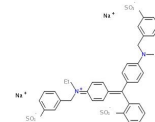
SED Adult mg/kg bw/day: 0.030	MoS - Adult 60kg: 22540.5	NO(A)EL mg/kg bw day: 695	
SED Child mg/kg bw/day: 0.110	MoS - Child 16.7kg: 6273.7	NOAEL test method:	reproductive study in rats
SED Baby mg/kg bw/day: 0.313	MoS - Baby 5.9kg: 2216.4		

Toxicological Summary:

A thoroughly investigated dye with low oral toxicity. Widely used in food (an FDA colour) and cosmetics. Unlikely to cause adverse effects when used in cosmetics.
The Cosmetics Regulation 1223/2009 has amended the use of this colour to be used in non oxidative hair dyes at a maximum concentration of 0.4%. This does not effect its use in other permitted cosmetic product types.

EFSA have reviewed Allura Red in its report published in 2009.
Toxicological findings:Acute Toxicity LD50 10000 (rat) and 5000 (dog) mg/kg bw, respectively
Allura Red has an ADI of 7 mg/kg bw/day and was based on a NOAEL of 695 mg/kg bw/day derived from a reproductive toxicity study in rats which revealed slight growth suppression observed mainly at the high test levels of 2595 mg/kg bw/day in F1 and F2 pups.

Scientific Opinion on the re-evaluation of Allura Red AC (E 129) as a food additive 1, EFSA Journal 2009; 7(11):1327



Chemical Structure:

Cosmetic Regulatory Summary:

EU Cosmetics Status: Approved colour all products
Saudi Cosmetics Status: Permitted colour field 1 All products
US Cosmetics Status: Generally, including cosmetics 74.101.

Canadian Cosmetics Status: Approved colour all products including eyes

Regulatory Summary:

EU Classification: Unclassified
GHS Classification: Unclassified
REACH Annex XVII controlled: Not Controlled
REACH SVHC Candidate List: Not Controlled

Systemic Exposure Dosage / Margin of Safety:

SED Adult mg/kg bw/day: 0.000	MoS - Adult 60kg: 3405405405.4	NO(A)EL mg/kg bw day: 630
SED Child mg/kg bw/day: 0.000	MoS - Child 16.7kg: 947837837.8	NOAEL test method: female rat, one generation chronic oral study
SED Baby mg/kg bw/day: 0.000	MoS - Baby 5.9kg: 334864864.8	

Toxicological Summary:

A dye thoroughly tested for use as a food additive (an FDA colour) and cosmetic dye. Unlikely to cause adverse effects when used in cosmetics. Permitted in US for use in all cosmetic products including those used near the eye. Unlikely to cause adverse effects at the typical concentrations used in cosmetics. Widely used and well accepted in cosmetic products. Canada cosmetic use; coal tar dye permitted in the area of the eye. The Cosmetics Directive 2009/36/EC has amended the use of this colour to be used in non oxidative hair dyes at a maximum concentration of 0.5%. This does not effect its use in other permitted cosmetic product types. Approved colour for use in finger paints EN71:7

CI 42090 (Acid Blue 9; CAS No. 3844-45-9) is also allowed as a food colorant in the EU with an ADI of 1-10 mg/kg bw/day. Acid Blue 9 has long history of use in a wide range of applications and it is "generally recognized as safe (GRAS)". Has low acute toxicity (Oral LD50 > 2000 mg/kg/d; rat, mouse). No data by the dermal and inhalation routes of exposure. Moderate skin irritation was reported in 40 of 207 volunteers in a testing of 5% aqueous solution but at 1%, it was non-irritating to skin in mice and rabbits. 10% aqueous solution of CI 42090 produced transient eye irritation in rabbits treated over a 4-week period and in HET-CAM test, only slight irritation was observed. Also, no irritation was reported in the neutral red uptake (NRU) assay on human keratinocytes at doses from 0.68 -10 mg/ml; the NRU-50 is reported > 10 mg/ml. In a local lymph node assay (LLNA), CI 42090 was found non-sensitising on dermal contact (stimulation indices < 3) and showed no potential for photosensitisation. Not mutagenic or genotoxic (bacterial, chromosome aberration, in vivo UDS assays). No significant toxicity effect was reported in oral administration of Acid Blue 9 at level of 4% in the diet (~ 2000 mg/kg bw/day) for 3 weeks in rats. Chronic toxicity studies have been conducted in rats and mice with Acid Blue 9. Mice were less sensitive with no adverse effects reported at the highest concentration of 9000 mg/kg bw/day (5% in diet). However, decreased in body weight and survival was seen in female rats dosed 2000 mg/kg bw/day (~ 2% in diet) and the NOAEL was concluded as 630 mg/kg bw/day (1% Acid Blue 9 in diet). The weight of evidence from available data indicates that it is not carcinogenic on oral administration. Acid Blue 9 was not teratogenic in rats and rabbits in doses up to 2000 mg/kg bw/day and 200 mg/kg bw/day, respectively. Reproductive toxicity was not reported with Acid Blue 9 in one-generation study (NOAEL: 1000 mg/kg bw/day; 2% in the diet). However, effects on offspring (bodyweights) and slight increase in the incidence of lung and kidney abnormalities in adult males were reported in a two-generation study and the LOAEL was determined to be 1000 mg/kg bw/day. The oral and percutaneous absorption of Acid Blue 9 is low (< 5%). An in vitro study on pig skin revealed maximum penetration rate of 35.5 µg/cm² within 24 hours (i.e. 0.60% of the applied dose) for a 0.5% product.

Note: In the absence of NO(A)EL data, the Margin of Safety (MoS) has not been calculated. Unless otherwise determined and in the absence of literature or other experimental data, a Dermal Absorption (DAP) of 100% is taken as the worst case scenario. NO(A)EL: No Observed Adverse Effect Level; MoS: Margin of Safety; SED Systemic Exposure Dosage; Calculation of Margin of Safety: MoS = NO(A)EL / SED

Reference for skin surface area, exposures and application quantities are derived from:

1. RIVM Report 320104001/2006
2. References cited in Dermal Sensitization Quantitative Risk Assessment (QRA) For Fragrance Ingredients, 2006 revision
3. Exposure factors handbook 2011
4. SCCS's Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation 8th Revision SCCS/1501/12
5. Colipa Data SCCNFP/0321/02
6. McNamara et al, Food Chem. Toxic; 2007, 45, 2086; Loretz et al, Food Chem. Toxic; 2008, 46, 1516
7. Body weights taken from Exposure factors handbook 2009 Update and mean values have been used unless specified otherwise
8. ConsExpo database
9. New default values for the spray model, RIVM, March 2010
10. Commission Implementing Decision of 25 November 2013 on Guidelines on Annex I to Regulation (EC) No 1223/2009 of the European Parliament and of the Council on cosmetic products (2013/674/EU)

This formulation has been assessed by Intertek in accordance with Part B, Annex I to Regulation (EC) No. 1223/2009 of the European Parliament and of the Council of 30 November 2009 on cosmetic products. The safety assessment is based upon the chemical specification and toxicological profile of the ingredients as supplied at the time of assessment and an assessment of the final cosmetic product. The supplier to this safety assessment is advised to ask for a new safety evaluation if any change in formulation occurs, change in raw materials used, abnormally high number of adverse events are recorded, changes in recommended uses or other circumstances that may affect the safety of this product.

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